PRIMER · FOR LOCAL POLICYMAKERS

Policymaker Primer

A starting point for local leaders. How data centers work, the local powers that govern them, their local impacts, and how to find out who the developer really is.

From the Grade My Data Center research team

What is a data center?

A data center is a building or part of a building that houses computer equipment to process, store, and transmit digital information. Every time you watch a video online, search the Internet, or use an AI tool, your request goes to a data center - the physical infrastructure for the Cloud.

Data centers range in size, generally characterized by power capacity (in megawatts or MW) and square footage. Small facilities called edge and micro data centers are often located in cities, close to people and customers. Medium-sized, co-location facilities have multiple tenants and are generally within suburban office parks. Hyperscale facilities - the largest - are vast campuses with multiple buildings often occupied by a single Big Tech occupant.

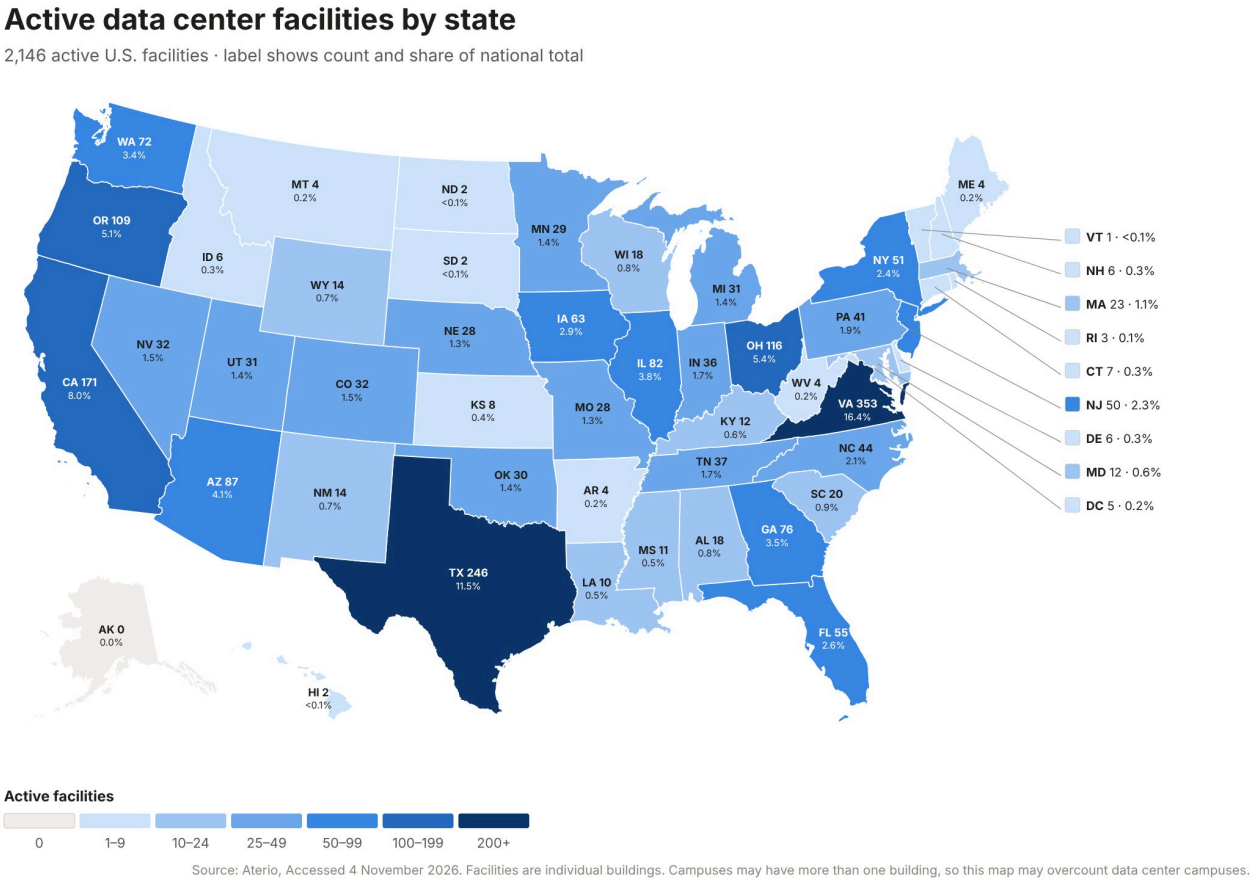
The computer equipment in data centers uses a lot of power to operate and puts off a lot of heat when it's running. Therefore, data centers may also need their own dedicated power and special building equipment, including air handlers, power generators, cooling systems, water storage, and even utility substations and nearby electricity transmission infrastructure. Some of this equipment will be built onsite with the data center; other equipment will need to be built nearby and sometimes in partnership with the local water or electric utility. Which equipment the data center uses and how often the equipment runs can have a big impact on local air pollution, water usage, greenhouse gas emissions, neighborhood noise, and light at night.

Data centers also require digital connectivity via fiber optic cables. These cables transmit data via light pulses, and because of their heavy weight, often dictate the design of the data center building. They also determine where data centers are sited, as developers have three primary requirements to build a data center: power availability through the local electric grid or onsite generation, fiber optic cable access and ease to lay, and land cost and availability.

Where are data centers?

No one knows the exact number of data centers in the U.S., but estimates range between 2,000 and 5,400 facilities. There is no single list. However, we do know there are data centers in every single state except Alaska. Virginia, Texas, and California have the most. Virginia alone has more than 15% of all U.S. data centers.

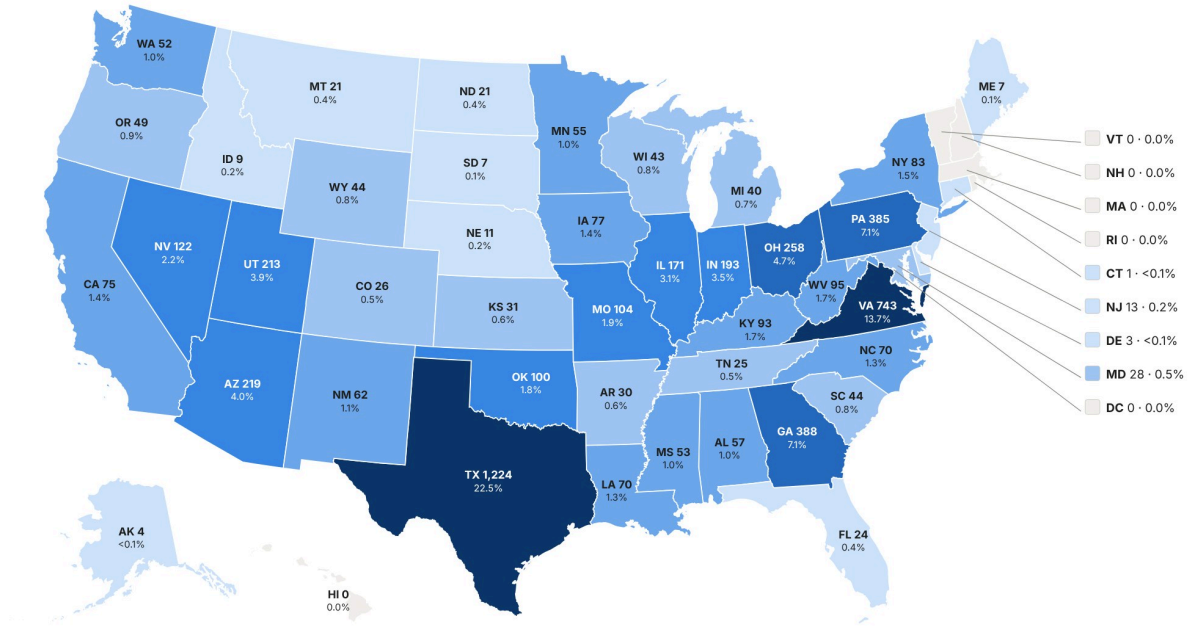
There are another 4,400 data centers proposed to be built soon. Many will go to Virginia, Texas, and California. Hundreds more are planned for states like Georgia, Pennsylvania, Arizona, Illinois, and Ohio.



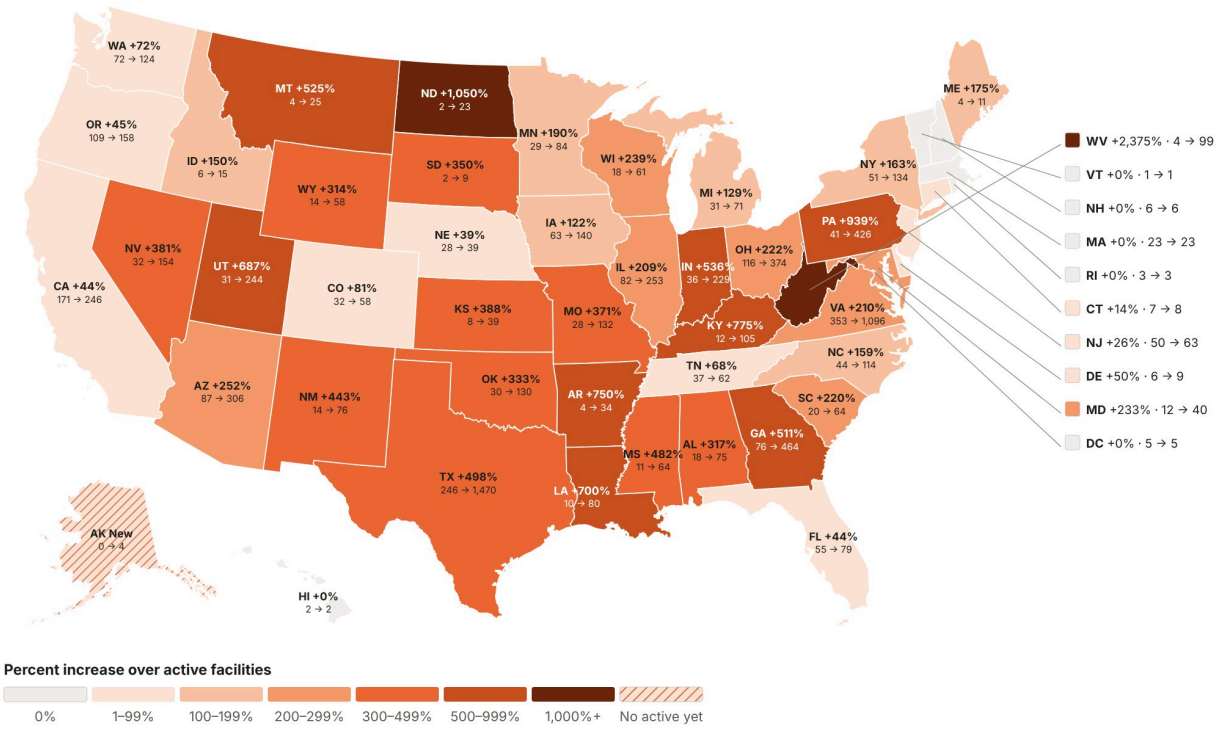
Source: Aterio, Accessed 4 November 2026. Facilities are individual buildings. Campuses may have more than one building, so this map may overcount data center campuses.

Under construction, delayed, and announced data center facilities by state

5,443 U.S. facilities: under construction (890), delayed (88), or announced (4,465) · label shows count and share of national total



Source: Aterio, Accessed 4 November 2026. Facilities are individual buildings. Campuses may have more than one building, so this map may overcount data center campuses.



Labels show percent increase and active facilities → active + under construction, delayed, and announced. Alaska has no active facilities (0 → 4).
Source: Aterio, Accessed 4 November 2026. Facilities are individual buildings. Campuses may have more than one building, so this map may overcount data center campuses.

What local regulatory powers govern data centers?

Local governments can choose to make development easy for data centers, including local regulations that allow for by-right development without additional approval or public input and, in some cases, offer tax breaks or even tax incentives. Or, they can wield local powers to impose restrictions on where and what type of data centers are proposed and gain additional benefits from their development through negotiated legal agreements or local policies.

Zoning

The critical tool local governments have to regulate data centers is zoning, or determining where data centers can be built, if at all, within a local jurisdiction. Although data centers have been around for a long time (since 1945), they are relatively new uses for the zoning code. This is because data centers are often large, windowless buildings with relatively few employees and little traffic going in-and-out of the facilities, apart from during construction. Most codes therefore treat data centers as industrial uses, warehousing and logistics, manufacturing, or sometimes even as offices. This can allow for data centers to be built "by-right," or without additional public input or local council approval.

Amendments to the zoning code classify "data centers" as its own use, with size tiers based on megawatts or square footage, rather than allowing projects to ride on generic warehouse or utility definitions. Furthermore, zoning amendments can attach operating standards like setbacks from homes and schools, decibel caps measured at the property line, shielded full-cutoff lighting, closed-loop or air-cooled cooling requirements, and water gallons-per-day caps, generator emissions tiers, "will-serve" letters requiring utilities to confirm capacity before a permit issues, and decommissioning plans backed by a surety bond - among other provisions.

Permitting

Depending on how and where they are built, data centers might need federal, state, and local permits. For local governments, a key permitting decision is which approval track data centers sit on, since routing them through a special use or conditional use permit preserves project-by-project approval and a public hearing. Administrative review, meaning review by the planning or building and safety department, does not.

Additionally, the permit application itself can be used to compel disclosure of peak MW demand, water source, projected gallons of water used per day, cooling technology, etc., since localities cannot regulate impacts they have not measured, and disclosure requirements face fewer legal challenges than substantive limits. Furthermore, conditions attached to a permit are enforceable and stay with the property if it is sold, so they could be written as specific and measurable standards and paired with phased approval, permit expiration dates, and other monitoring measures.

Moratorium

Some jurisdictions might decide that they want a temporary pause on accepting or approving data center applications (a moratorium) or even a ban. Usually six months to a year, a moratorium allows local government staff to study impacts and write permanent rules governing data centers. To survive a legal challenge, the record should show written findings that the pause exists to enable a specific study, a fixed end date, neutral application to all similar properties, and grandfathering of pending applications. The key thing to understand is that a moratorium buys time and nothing else; the ordinance written during the pause is what actually governs afterward. If allowed by the state government, local governments might also pursue a total ban on data centers, where no data centers - or no data centers above a certain size - are allowed to be built.

Taxes

The types of taxes local governments can levy vary widely by state in the U.S. However, for data centers, local governments can receive most of their fiscal benefit from real property tax on data center buildings and personal property tax on computer equipment. Local governments can charge real property tax rates specific to data center facilities, and reassess properties based on higher values of data centers when there is a change of use. Servers, or computer equipment, depreciate quickly, so an equipment tax base takes advantage of companies' reinvesting.

Retirement & decommissioning policies

Conditional-use permits (CUPs) can require a decommissioning plan backed by a bond, also known as a reclamation bond or site restoration bond, which ensures the site is "safely dismantled at the end of its useful life" and restored to its original condition. The mechanism is still rare for data centers but well established in wind, solar, and mining permitting.

Community benefits ordinances & agreements

A community benefits ordinances is a local policy that requires developers of projects of a certain size or type to sign a community benefits agreement (CBA) or other type of legally binding contract with local residents, groups, or the local government itself. The CBA typically covers direct payments, infrastructure funding, local hiring and wage standards, noise and water caps, and annual public reporting. CBA terms carry the most weight when settled before the rezoning or permit vote and written as measurable commitments, and also supplement a zoning ordinance rather than substituting for one. For example, the City of Lancaster, PA, negotiated a **CBA** with the Lancaster AI Hub for a $20.25 million direct payment, plus a 20,000 gallon-per-day water cap.

Nondisclosure agreements

Some developers might request that local elected officials or government staff sign nondisclosure agreements (NDA), which limits or forbids them from speaking with the public about not just the proposed data center development, but also the existence of the NDA itself. This is highly problematic, and signing NDAs should be avoided at all costs.

What are the local impacts of data centers?

Energy & electricity

The servers in data centers consume a lot of energy and produce a lot of waste heat. A single hyperscale data center can use more than 100 megawatts (MW) of power, roughly the same amount of electricity drawn by 100,000 homes at any given moment, or by 3,000 electric vehicles charging simultaneously. This power usually comes from the local grid, the same electric utility that delivers power to your jurisdiction. Some data centers generate their own power on-site.

Heat

When thousands of servers run at the same time, they give off heat. Most of this heat is cooled down by large air conditioners inside the building before it is released outside. However, research shows that letting this warm air out can still raise the local outdoor temperature, creating a warm pocket of air called a heat island.

Water usage

Data centers can use water in two ways. First, they use it to cool down hot servers. Depending on the cooling system they use, a data center might use millions of gallons of water per day (like filling multiple swimming pools) or they might use a closed-loop system that recycles water so it only uses as much as a toilet flush. Second, a lot of water is used at the power plants that make the facility's electricity, even if those power sources are onsite at the data center campus. This water generally comes directly from the same local water utilities that supply your jurisdiction.

Noise & light

Data centers never cease operations. They run 24 hours a day, 365 days a year. Because of this, their giant cooling fans and electrical equipment make constant, low frequency humming noise. In Prince William County, Virginia, residents near data centers have reported exterior noise routinely exceeding 60 dBA, well above the 45 to 55 dBA nighttime limits set by most residential noise ordinances. Compliance measurements are typically taken in A-weighted decibels (dBA), which discount low-frequency sound; a facility can therefore meet code while still producing the persistent hum neighbors describe. Neighbors say it sounds like a refrigerator or hairdryer that never turns off. To keep the buildings safe, bright security lights stay on all night.

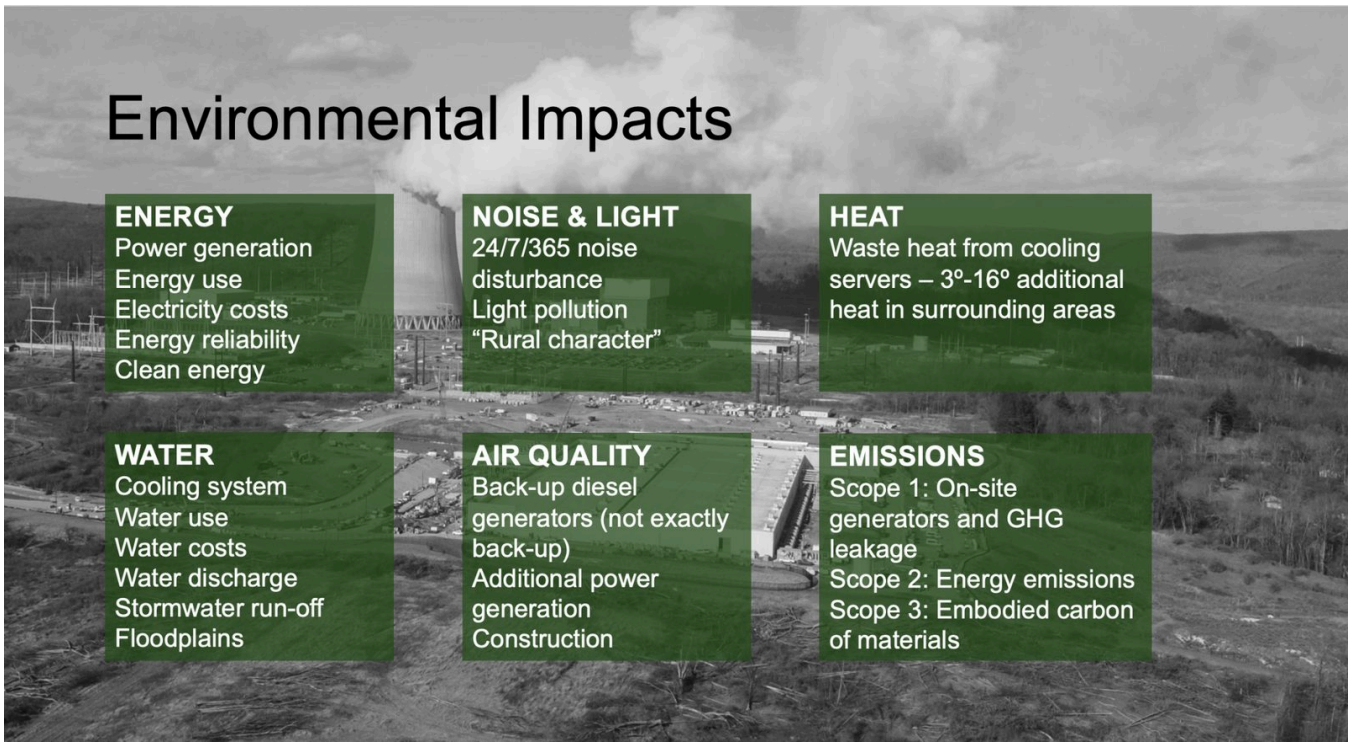
Jobs

Data centers bring different kinds of jobs. During construction, they create hundreds or thousands of temporary jobs for builders. Because hyperscale campuses are so big, construction can last for several years. Once the building is finished, however, it is mostly run by computers. A finished hyperscale data center only needs a couple hundred permanent workers to keep it running. Smaller data centers need even fewer workers. Often, job incentives cited by data center development companies are merely temporary boosts to the local economy, not lasting prosperity.

How do I figure out who the data center developer is?

The name on the application is frequently not the name of the company that will run the facility. Most projects are filed by a limited liability company created for that single site, sometimes under a code name, and that entity is often a joint venture between a developer, an infrastructure or private credit fund putting up the capital, and sometimes the eventual tenant. In this arrangement the roles are split: an investor finances the project, a project owner builds and holds the real estate, and a tenant signs a long-term lease, often fifteen to twenty years and frequently signed before construction begins.

The exception is a large technology company building and owning its campus directly - Meta, Google, Amazon Web Services, and Microsoft all do this - and those projects are usually announced under the company's own name. Because the applicant in front of you may never operate the site and may not own it in five years, conditions should be written to bind the property and any successor owner, and disclosure of the parent company, intended tenant, and projected power and water demand can be required as a condition of consideration.



Environmental impacts



Social impacts



Design impacts

Glossary

A-weighted decibels (dBA)

A common way to measure loudness that matches how people hear, but counts low, rumbling sounds less. A data center can meet a dBA noise limit and still make a steady hum.

By-right development

A project the zoning code already allows, so staff can approve it without a public hearing or a council vote.

Campus

A group of buildings that all belong to the same data center, sitting on the same piece of land and working together.

Closed-loop system

A way of cooling computers that reuses the same water over and over. It uses much less water than systems that drain the water away after one use.

Co-location data center

A medium-sized data center that rents space to several companies, often in a suburban office park.

Community benefits agreement (CBA)

A legally binding contract between a developer and local residents, groups, or the local government that spells out what the community will get, such as payments, local hiring, or limits on water use and noise.

Community benefits ordinance

A local law that requires developers of certain large projects to sign a community benefits agreement.

Conditional use permit (CUP)

Approval for a use that is allowed only under certain conditions, decided project by project and usually after a public hearing. Also called a special use permit. Its conditions stay with the property if it is sold.

Decommissioning plan

A plan for taking a facility apart and restoring the site when it closes, often backed by a bond.

Edge or micro data center

A small data center, often in a city, placed close to the people and businesses it serves.

Electric grid

The system of power lines and equipment that carries electricity from power plants to homes, schools, and businesses.

Electric utility

The company that supplies electricity to homes and businesses in an area.

Hyperscale data center

A very large data center campus that can hold thousands or even hundreds of thousands of servers, usually owned and used by a big tech company like Amazon, Google, Microsoft, or Meta.

Megawatt (MW)

A unit used to measure electricity. One megawatt can power about 1,000 average homes. Data center size is often described in megawatts.

Moratorium

A temporary pause, usually six months to a year, on accepting or approving certain applications while a local government studies impacts and writes permanent rules.

Nondisclosure agreement (NDA)

A contract that bars the people who sign it from sharing certain information. Some developers ask local officials or staff to sign one.

Personal property tax

A tax on business equipment, such as the servers and computer equipment inside a data center.

Project company

A limited liability company (LLC) set up for a single project. Its name is often the one on applications, even when a different company will own or run the data center.

Real property tax

A tax on land and buildings.

Reclamation bond

Money a developer sets aside, also called a site restoration bond, to pay for removing a facility and restoring the land if the owner does not.

Server

A powerful computer that stores information and sends it to other devices. Data centers are filled with rows of servers.

Setback

A required minimum distance between a building or equipment and a property line, homes, schools, or other uses.

Substation

Electrical equipment that changes the voltage of power from transmission lines so it can be delivered to homes or large customers like data centers.

Tenant

The company that leases and uses a data center, often under a lease of fifteen to twenty years.

Will-serve letter

A letter from a utility confirming that it has the capacity to provide power or water to a project.

Zoning

Local rules that decide what kinds of buildings and uses are allowed in each part of a town or county.

Zoning text amendment

A change to the wording of the zoning code, for example to define data centers as their own use or to set where they may be built.

Glossary built with the assistance of Claude (Anthropic).

Resources

Stop Dirty Data Centers

The NAACP's project creates a proactive approach for local leaders, advocates, and organizers to understand and act on community, transparency, policy, and power issues. They offer templates for local policymakers and community members on the subjects of **negotiation**, **community benefit agreements**, and **cost recovery support agreements**. Additionally, the NAACP provides informational sources about **advocacy in action**, **AI and the environment**, and **impacts** of data centers.

Data Center Laws and Regulations

This **We Build Progress** tracker recounts state and local level data center regulations and policies. It includes the category and status of the policy, links to news about it, and which state and/or county and locality it's coming from.

Climate Mayors

This resource provides data center issue briefs for local policymakers on the subjects of water, air quality, electricity, the economy, and extreme heat. Additionally, the **Data Centers and the Climate Landscape: An Actionable Resource For US Mayors** features potential community and climate safeguards, a deep-dive into all things data centers, and an overview of city policy levers as well as case studies.

Loudoun County, Virginia: Data Center Capital of the World, "A Strategy for a Changing Paradigm"

This white paper is written in 2024 and revised in 2026 by Mike Turner, Vice Chair of the Loudoun County Board of Supervisors. This 46-page document is specific to Loudoun County, but offers insight into a local policymakers' perspective on managing data centers and includes **15 Best Practices for Communities Considering Data Centers**.

National Association of Counties: Data Center Resource Hub

A dynamic collection of strategies and resources to strengthen localities' approach and oversight of data centers. This resource includes webinars, primers, conferences, case studies, and more resources from partner organizations, all customized to a local policymaker lens.